

Dingwei Guo

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SUMMARY OF HANDS-ON SKILLS

- **Experimentation & Causal ML:** A/B testing, survey & experiment methodology, Difference-in-Differences, Regression Discontinuity, Instrumental Variables, Double ML, Causal/IV Forest, matching, and weighting.
- **Machine Learning & Structural Modeling:** supervised learning, ensemble and tree methods, regularization, heterogeneous treatment-effect modeling, structural modeling, and model validation.
- **Bayesian, Computational & Forecasting Methods:** Bayesian inference, hierarchical modeling, Monte Carlo methods, numerical optimization, time-series forecasting, surrogate modeling, and causal forecasting.
- **Programming & Data Systems:** Python, SQL, R, PySpark, scikit-learn, LightGBM, XGBoost, TensorFlow, AWS, SageMaker, Airflow, Docker, MLflow, and Git for scalable modeling and productionized analytics.

EDUCATION

- Ph.D. in Economics, **University of North Carolina at Chapel Hill** 2027
- M.S. in Economics, **University of Wisconsin-Madison** 2022
- B.A. in Economics, **Xi'an Jiaotong University** 2021

INDUSTRY & RESEARCH EXPERIENCE

- **Amazon | Prime Video Marketing Measurement | Economist Intern** May 2026 - Aug 2026
 - Built a causal measurement system for Prime Video advertising using 10M+ observations, combining Double ML, Regression Discontinuity, A/B testing, and high-dimensional feature engineering in SQL, Python, and AWS.
 - Applied Double ML, causal forest, and IV forest methods to estimate heterogeneous treatment effects across content verticals, customer segments, and post-exposure horizons.
 - Productionized a scalable, automated data and modeling pipeline on AWS using SQL, PySpark, and Python for causal measurement, model validation, and recurring campaign monitoring.
 - Integrated experimental and quasi-experimental causal estimates into a broader Bayesian hierarchical multi-channel framework to calibrate observational signals and improve channel-level contribution estimates.
 - Translated model outputs into recommendations for senior product, engineering, and marketing stakeholders, directly supporting decisions on next-year ad targeting, frequency, campaign duration, and measurement strategy.
- **Experimentation, Survey Methodology, Structural Model, and Causal Forecasting | Ph.D. [Research](#)**
 - Designed randomized experiments and estimate structural choice models using moment-based methods, Monte Carlo integration, and numerical optimization to recover consumer demand and counterfactual participation.
 - Compared alternative stated-preference elicitation formats and quantified how survey design affects willingness-to-pay estimates.
 - Trained regularized regression, random forest, and gradient boosting surrogate models with cross-validation to extrapolate short-run experimental responses to longer-run outcomes.
 - Published and forthcoming research in *Frontiers in Energy Research*, *Energy Strategy Reviews*, and *Environment and Development Economics*.
- **Quasi-Experimental Policy Evaluation | Ph.D. [Research](#)**
 - Led five quasi-experimental studies on the labor-market and intergenerational effects of large-scale fiscal, labor, and social-assistance policies.
 - Built longitudinal panel datasets by integrating household microdata, administrative records, geospatial indicators, and unstructured documents using web scraping, text parsing, geospatial matching, and record linkage.
 - Applied staggered Difference-in-Differences, event study, instrumental variables, matching/weighting, and heterogeneous treatment-effect estimation to recover policy impacts from observational data.
 - Published and forthcoming research in *Journal of Development Effectiveness* and *Agricultural and Food Economics*.

AWARDS

- 2026 Best Graduate Research Award in Economics UNC Chapel Hill
- Graduate Fellowship UNC Chapel Hill